

DRAWING LIST FOR TEST PACKAGE

SLD-661

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	661	5FA	100A-WW-SLD-ST1-N-8531-01	1 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
2	661	5FA	100A-WW-SLD-ST1-N-8532-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	
3	661	5FA	40A-IW-ST1-C-9566-01	1 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
4	661	5FA	40A-IW-ST1-C-9566-02	2 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
5	661	5FA	100A-WW-SLD-ST1-N-8531-02	2 OF 2	0	CJ-M15W14-PR-A20-000108	1A	
6	661	5FA	100A-WW-SLD-ST1-N-8533-01	1 OF 1	0	CJ-M15W14-PR-A20-000108	1A	

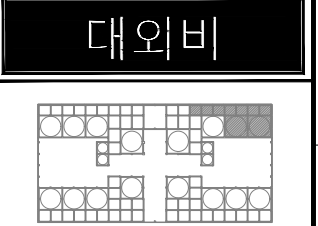
ITEM NO.	W14-T-4127 A
SERVICE	CIF 2'nd NEUTRALIZATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 4.925 x 8.15 (EH 6.5)
CAPA.	m3 (220.8 (EH 176.1))
HRT	min (34.9)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4128 A
SERVICE	CIF 2'nd COAGULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.5 x 4.925 x 8.15 (EH 6.5)
CAPA.	m3 (220.8 (EH 176.1))
HRT	min (34.9)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4129 A
SERVICE	CIF 2'nd FLOCCULATION TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 5.7 x 4.925 x 8.15 (EH 6.5)
CAPA.	m3 (228.8 (EH 182.5))
HRT	min (36.1)
MAT'L	P.C + FRP

ITEM NO.	W14-T-4130 A
SERVICE	CIF 2'nd SETTLING TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H m 11.7 x 10.825 x 8.15 (EH 6.5)
CAPA.	m3 (1032.2 (EH 823.2))
HRT	hr 2.7
MAT'L	P.C + FRP

ITEM NO.	W14-LP-4130 A
SERVICE	LAMELLAR PLATE
Q'TY	1
CAPA.	W x L x H m BY VENDOR
SURFACE LOAD	20 m3/m2.d 이하 (Qmax : 6,450m3/d)
MAT'L	ABS+SS304



KEY PLAN (5F)

- NOTE
1. BAFFLE 재질 : FRP
BAFFLE 상부에 0.02 m2 이상 AIR HOLE 설치.
 2. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
 3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
 4. ANHOLE : 2개 이상 설치한다.
ANHOLE COVER : FRP
 5. LOTO 시 CLOSE 되는 배관에는 LOTO 설치.
 6. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
 7. AGITATOR MAT'L 표기 : SHAFT/BLADE
 8. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 최소 길이로 반영 할 것.
(SLAB으로부터 100mm 떨어질 것)
 9. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
 10. 위에 것수등의 상세사항은 추후 VENDOR에서 결정한다.
 11. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET/OUTLET NOZZLE SIZE 확정
 12. FLUSHING 용 WATER / AIR 배관은 45도 서선으로 연결할 것.
 13. SLUDGE REMOVAL PUMP 또는 배관은 PUMP SUCTION 속의 TANK 수위보다 HIGH POINT 형성.
 14. DELETED.
 15. DELETED.
 16. DELETED.
 17. DELETED.
 18. DELETED.
 19. 시반 형소 및 유지관리시설을 PACKAGE VENDOR 공급범위에 포함할 것.
 20. SLUDGE PUMP는 TIMER에 의해 운전됨.
 21. DELETED.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
375	실명	날짜
375	DESCRIPTION	DATE

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DRAWN BY
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CHECKED BY
승 인 박 동 명
APPROVED BY
제 작 사

SK 에코플랜트

공 사 명
PROJECT TITLE

M15X Ph4 WWT Project

도면명
DWG. TITLE

PIPING & INSTRUMENT DIAGRAM

CIF 2'nd COAGULATION & SETTLING TANK A (1/2)

속 척
SCALE NONE

날 짜
DATE 2025. 08. 14

PROJECT NO. 22330D REV. A

도면번호
SHEET NO. CJ-M15W14-PR-A20-000108 1A

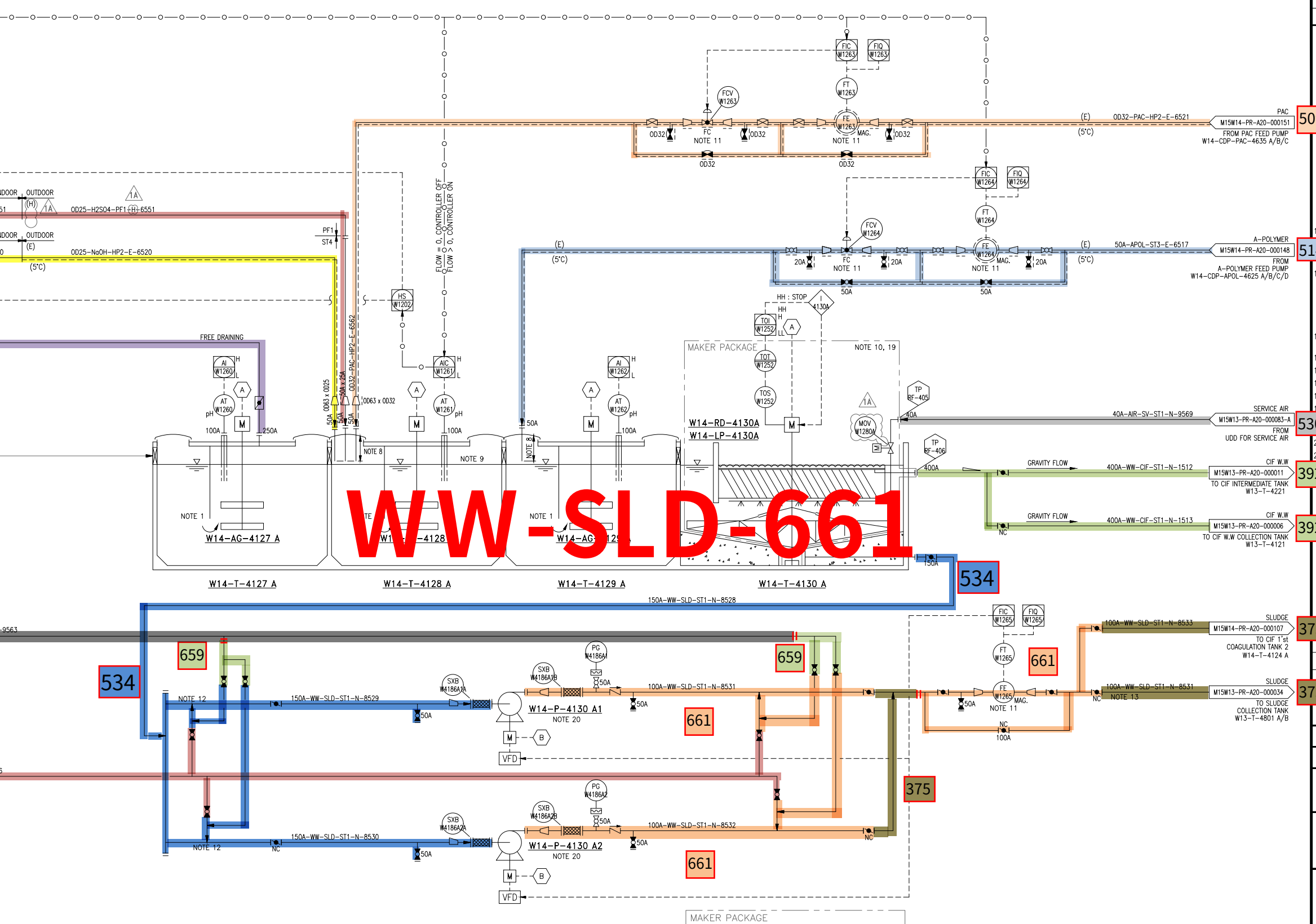
ITEM NO.	W14-AG-4127 A
SERVICE	CIF 2'nd NEUTRALIZATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (30)
MAT'L	STS316L/STS316L
POWER	kw (15)

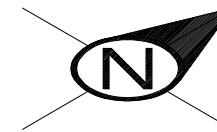
ITEM NO.	W14-AG-4128 A
SERVICE	CIF 2'nd COAGULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (11)

ITEM NO.	W14-AG-4129 A
SERVICE	CIF 2'nd FLOCCULATION AGITATOR
Q'TY	1
TYPE	HYDRO PADDLE NOTE 15
SPEED	rpm (25)
MAT'L	STS316L/STS316L
POWER	kw (17.5)

ITEM NO.	W14-RD-4130 A
SERVICE	CIF 2'nd REDUCER DRIVER
Q'TY	1
TYPE	CYCLO REDUCER
CAPA.	m3/hr.ea x mH (0.067)
MAT'L	STS304/STS304
POWER	kw (2.2)

ITEM NO.	W14-P-4130 A1/A2
SERVICE	CIF 2'nd SLUDGE RETURN PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL (WVF), VORTEX
CAPA.	m3/hr.ea x mH 50 x 30 (0.29MPa)
MAT'L	SSC13/SSC19
POWER	kw (30)





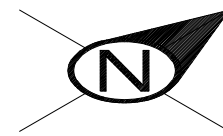
1. (*) FIELD ADJUST

[illegible]

CLIENT		 2	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WW—SLD		22330D	
LINE NO.	100A-WW-SLD-ST1-N-8532	SH'T NO.	REV.
		01 / 01	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1												
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	10	60	UNIT 5FA						
			OPERATING	2.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050004-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000108						
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

[illegible]



<NOTE>
1. (*) FIELD ADJUST

CLIENT  3

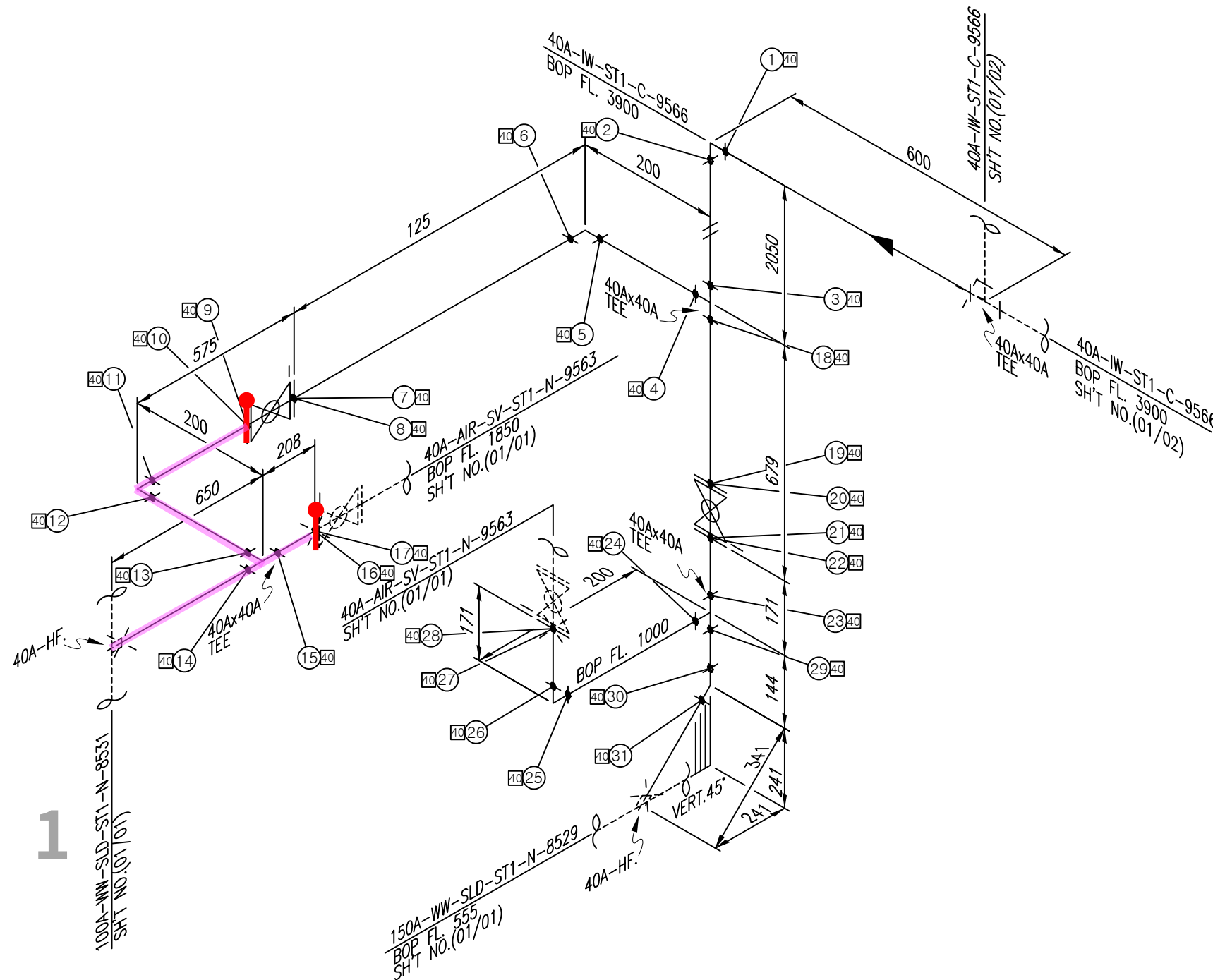
INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	13.5	60	UNIT 5FA						
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050004-D		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000108						
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

CONTRACTOR 		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
IW		22330D	
LINE NO.	40A-IW-ST1-C-9566	SH'T NO. 01 / 02	REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	40A	7.0M	PIPE, STS 304TP, ERW, PE – SCH 10S
E90	40A	4EA	ELBOW 90 DEG, STSF304, SW – SCH 10S
E45	40A	1 EA	ELBOW 45 DEG, STSF304, SW – SCH 10S
T	40A	3EA	TEE EQUAL, STSF304, SW – SCH 10S
VB	40A	2EA	BALL VALVE, KS 10K, RF FLG'D, BODY–STSF304 /
			TRIM–STSF304 / SEAT–PTFE, FB, LEVER
F	40A	6EA	FLG SO, KS 10K, RF, STS304
G	40A	4EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
B	40A	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
IN	25tX40A	7.0M	관 보온재(고무발포)



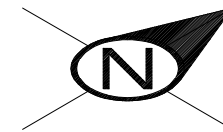
<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	C	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						CONTRACTOR	PROJECT NAME		
	THICK.	25	DESIGN	13.5	60	UNIT	5FA							M15X Ph4 Project		
			OPERATING	10	AMB	REF. PLAN DWG.							COMMODITY	JOB NO.		
						CJ-M15W14-PI-A30-050004-D			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	22330D		
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000108								40A-IW-ST1-C-9566	02 / 02	0

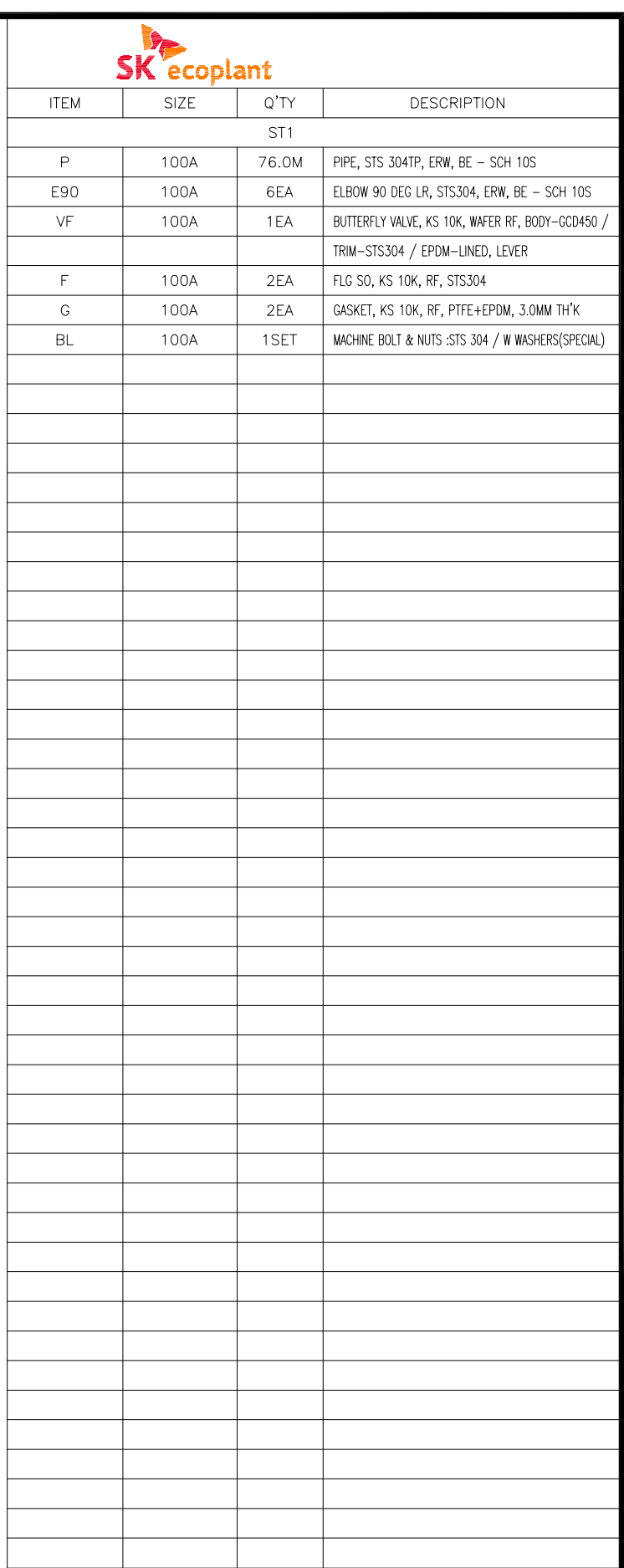
영고: 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단으로 사용, 복제, 제3자에게 공개, 누설, 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



<NOTE>
1. (*) FIELD ADJUST

CLIENT  5

정고 : 본 문건은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



CLIENT  6

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.